

Dynamic hashing provides a mechanism in which data buckets are added and removed dynamically and on-demand. Dynamic hashing is also known as extended hashing.

24. Deadlock

Deadlock is a situation which occurs when two transactions wait on a resource which is locked or other transaction holds. Deadlocks can be prevented by making all the transactions acquire all the locks at the same instance of time. So, once a deadlock occurs, the only way to cure it is to abort one of the transactions and remove the partially completed work.

25. Differences between an exclusive lock and a shared lock

Exclusive lock	Shared lock
An exclusive lock is a lock on a data item when a transaction is about to perform the write operation.	A shared lock allows more than one transaction to read the data items.

SQL

1. SQL

Structured Query Language is the core of the relational database which is used for accessing and managing the databases. This language is used to manipulate and retrieve data from a structured data format in the form of tables and holds relationships between those tables. So, SQL is used to communicate with the database.

2. Sub-queries

A subquery is a query inside another query where a query is defined to retrieve data or information back from the database. In a subquery, the outer query is called the main query whereas the inner query is called subquery. Subqueries are always executed first and the result of the subquery is passed on to the main query. It can be nested inside a SELECT, UPDATE or any other query. A subquery can also use any comparison operators such as $>$, $<$ or $=$.

3. Data Definition Language(DDL)

DDL changes the structure of the table like creating a table, deleting a table, altering a table. All the commands of DDL are auto-committed that means it permanently saves all the changes in the database.

DDL Commands

1. Create - It is used to create a new database or table.

```
CREATE TABLE TABLE_NAME (COLUMN1 DATATYPE, COLUMN2 DATATYPE,...);
```

2. Alter - It is used to alter the structure of the database like adding, renaming, dropping and changing the datatype of column.

```
ALTER TABLE TABLE_NAME ADD COLUMN_NAME DATATYPE;
```

3. Truncate - It is used to delete all the rows from the table and free the space containing the table.

```
TRUNCATE TABLE TABLE_NAME;
```

4. Drop - It is used to delete both the structure and record stored in the table.

```
DROP TABLE TABLE_NAME;
```

4. Data Manipulation Language(DML)

DML commands are used for manipulating the data stored in the table. It is responsible for all forms of changes in the database. The command of DML is not auto-committed, which means it can't permanently save all the changes in the database. They can be rolled back.

DML Commands

1. Insert - It is used to insert data into the row of a table.

```
INSERT INTO TABLE_NAME VALUES(VALUE1,VALUE2,VALUE3,...);
```

2. Update - This command is used to update or modify the value of a column in the table.

```
UPDATE TABLE_NAME SET COLUMN1=VALUE1,COLUMN2=VALUE2 WHERE  
CONDITION;
```

3. Delete - It is used to remove one or more rows from a table.

```
DELETE FROM TABLE_NAME WHERE CONDITION;
```

alter table employee add dept varchar(10);
→ update employee set dept = 'HR'
where employee_id = 1;
→ update employee set dept = 'IT'
where employee_id = 2;
→ update employee set dept = 'FIN'
where employee_id = 3;

5. Transaction Control Language(TCL)

These commands are used to keep a check on other commands and their effect on the database. These commands can annul changes made by other commands by rolling the data back to its original state. It can also make any temporary changes permanent.

TCL Commands

1. **Commit** - Commit command is used to save all the transactions to the database.

COMMIT;

2. **Rollback** - Rollback command is used to undo transactions that have not already been saved to the database.

ROLLBACK;

3. **Savepoint** - It is used to roll the transaction back to a certain point without rolling back the entire transaction.

SAVEPOINT SAVEPOINT_NAME;

6. Data Control Language(DCL)

DCL commands are used to grant and take back authority from any database user.

DCL Commands

- 1. **Grant** - It is used to give user access privileges to a database.
- 2. **Revoke** - It is used to take back permissions from the user.

7. Data Query Language(DQL)

DQL is used to fetch the data from the database.

DQL Command

1. **Select** - It is used to select the attribute based on the condition described by WHERE clause.

SELECT COLUMN1, COLUMN2 FROM TABLE_NAME;

8. Difference between Drop, Truncate, Delete

Drop	Truncate	Delete
Used to delete a database, table or a view	Used to delete all rows from a table	Used to delete a row in the table
Deletes the full structure of the table	Preserves the structure of the table	Deletes the structure of the row from a table

alter table employee
update employee set dept =
where employee-id =

9. Difference between WHERE and HAVING clause

WHERE	HAVING
WHERE clause works on row's data and cannot be used with aggregates	HAVING clause works on aggregated data
Used only with SELECT statement	Used in a GROUP BY clause
It is called pre filter	It is called post filter

10. LIKE clause

LIKE clause is used in the condition in SQL query with the WHERE clause. LIKE clause compares data with an expression using wildcard operators to match patterns given in the condition.

There are two wildcard operators that are used in the LIKE clause.

- **Percent sign %:** represents zero, one or more than one character.
- **Underscore sign _:** represents only a single character.

11. Order by clause

Order by clause is used with a SELECT statement for arranging retrieved data in sorted order. The Order by clause by default sorts the retrieved data in ascending order. To sort the data in descending order DESC keyword is used with Order by clause.

```
SELECT column-list|* FROM table-name ORDER BY ASC | DESC;
```

12. Group by clause

Group by clause is used to group the results of a SELECT query based on one or more columns. It is also used with SQL functions to group the result from one or more tables.

13. Having clause

Having clause is used with SQL Queries to give more precise condition for a statement. It is used to mention condition in Group by based SQL queries, just like WHERE clause is used with SELECT query.

14. Distinct

The distinct keyword is used with the SELECT statement to retrieve unique values from the table. Distinct removes all the duplicate records while retrieving records from any table in the database.

```
SELECT DISTINCT column-name FROM table-name;
```

15. SQL Joins

A JOIN clause is used to combine rows from two or more tables, based on related column between them. It is used to merge two tables or retrieve data from there.

16. Cross Join or Cartesian Product

This type of JOIN returns the cartesian product of rows from the tables in Join. It will return a table which consists of records which combine each row from the first table with each row of the second table. If there is no condition specified then it returns all possible pairing of rows from both the tables whether they are matched or unmatched.

```
SELECT column-name-list FROM table-name1 CROSS JOIN table-name2;
```

17. Inner Join

This is a simple JOIN in which the result is based on matched data as per the equality condition specified in the SQL query.

```
SELECT column-name-list FROM  
table-name1 INNER JOIN table-name2  
WHERE table-name1.column-name = table-name2.column-name;
```

18. Natural Join

Natural Join is a type of Inner join which is based on columns having the same name and same datatype present in both the tables to be joined.

```
SELECT * FROM table-name1 NATURAL JOIN table-name2;
```

19. Outer Join

Outer Join is based on both matched and unmatched data. Outer Joins subdivide further into,

Left Outer Join

The left outer join returns a resultset table with the matched data from the two tables and then the remaining rows of the left table and null from the right table's columns.

```
SELECT column-name-list FROM  
table-name1 LEFT OUTER JOIN table-name2  
ON table-name1.column-name = table-name2.column-name;
```

Right Outer Join

The right outer join returns a resultset table with the matched data from the two tables being joined, then the remaining rows of the right table and null for the remaining left table's columns.

```
SELECT column-name-list FROM  
table-name1 RIGHT OUTER JOIN table-name2  
ON table-name1.column-name = table-name2.column-name;
```

Full Outer Join

The full outer join returns a resultset table with the matched data of two tables then remaining rows of both left table and then the right table.

```
SELECT column-name-list FROM  
table-name1 FULL OUTER JOIN table-name2  
ON table-name1.column-name = table-name2.column-name;
```

20. Difference between Natural Join and Inner Join

Natural Join	Inner Join
Natural Join joins two tables based on the same attribute name and datatypes.	Inner join joins two tables on the basis of the column which is explicitly specified in the ON clause.
The resulting table will contain all the attributes of both the tables but keep only one copy of each common column.	The resulting table will contain all the attributes of both the tables including duplicate columns also.

21. Difference between Inner Join and Outer Join

Inner Join	Outer Join
Returns combined tuple between two or more tables.	Returns the combined tuple from a specified table even when the join condition fails.
When attributes are not common then it will return nothing.	It does not depend upon the common attributes. If the attribute is blank then NULL is placed there.

22. Difference between Join and Union

Join	Union
Combines data from many tables based on a matched condition between them.	Combines the result set of two or more select statements.
It combines data into new columns.	It combines data into new rows.
Number of columns selected from each table may not be same.	Number of columns selected from each table should be same.
It may not return distinct columns.	It returns distinct rows.

23. Set Operations in SQL

SQL supports few Set operations which can be performed on the table data. These are used to get meaningful results from data stored in the table, under different special conditions.

The different Set Operations are

- Union
- Union All
- Minus
- Intersect

to database vjaya;
vjaya;
table employee (
employee-id ~~integer~~);

alter table
update

29. Scalar Functions

Scalar functions return a single value from an input.

Following are some Aggregate functions

- UCASE()
- LCASE()
- MID()
- ROUND()

30. SQL Constraints

SQL Constraints are rules used to limit the type of data that can go into a table, to maintain the accuracy and integrity of the data inside the table.

Constraints can be divided into the following two types:

1. **Column level constraints:** Limits only column data.
2. **Table level constraints:** Limits whole table data.

Constraints are used to make sure that the integrity of data is maintained in the database. Following are the most used constraints that can be applied to a table.

1. NOT NULL
2. UNIQUE
3. PRIMARY KEY
4. FOREIGN KEY
5. CHECK
6. DEFAULT